

Programming Fundamental

Lab 4

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BSCS-1C

Task 1: Input and Display User Information

Objective

To write a C program that takes the user's age, height, and first letter of their name as input and displays the entered information.

Source Code

```c
#include <stdio.h>

int main() {
    int age;
    float height;
    char initial;

    printf("Enter your age: ");
    scanf("%d", &age);

    printf("Enter your height in feet: ");
    scanf("%f", &height);

    printf("Enter the first letter of your name: ");
    scanf(" %c", &initial);

    printf("\nYour Information:\n");
    printf("Age: %d\n", age);
    printf("Height: %.2f feet\n", height);
    printf("First Letter: %c\n", initial);

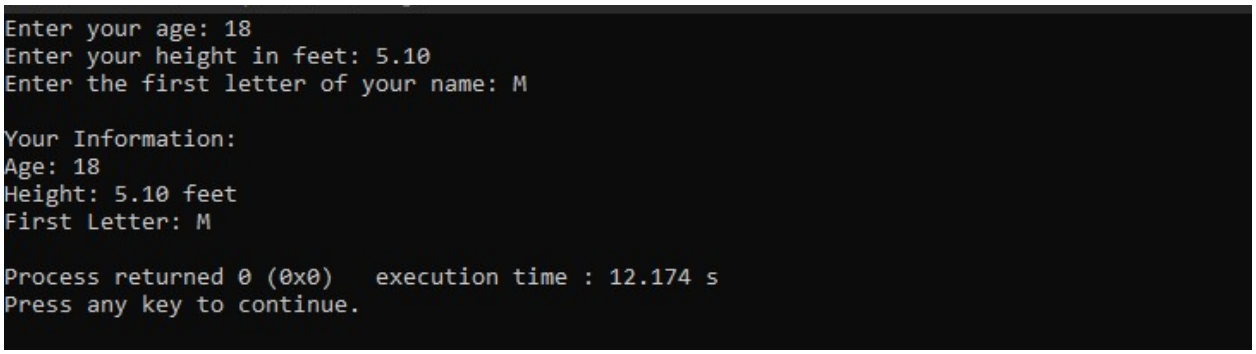
    return 0;
}

```

Explanation

- `int age` stores the user's age as a whole number.
- `float height` stores the user's height as a decimal number.
- `char initial` stores the first letter of the user's name.
- `scanf()` takes input from the user.
- `printf()` displays the entered values.
- `return 0` indicates that the program has ended successfully.

Sample Output



Result

The program successfully takes the user's age, height, and first letter of their name and displays the entered values.

Task 2:Swap Values of Two Variables

Objective

To write a C program that swaps the values of two integer variables using a temporary variable.

Source code

```c
#include <stdio.h>

int main() {
    int a = 10;
    int b = 20;
    int temp;

    printf("Before swapping:\n");
    printf("a = %d\n", a);
    printf("b = %d\n", b);

    temp = a;
    a = b;
    b = temp;

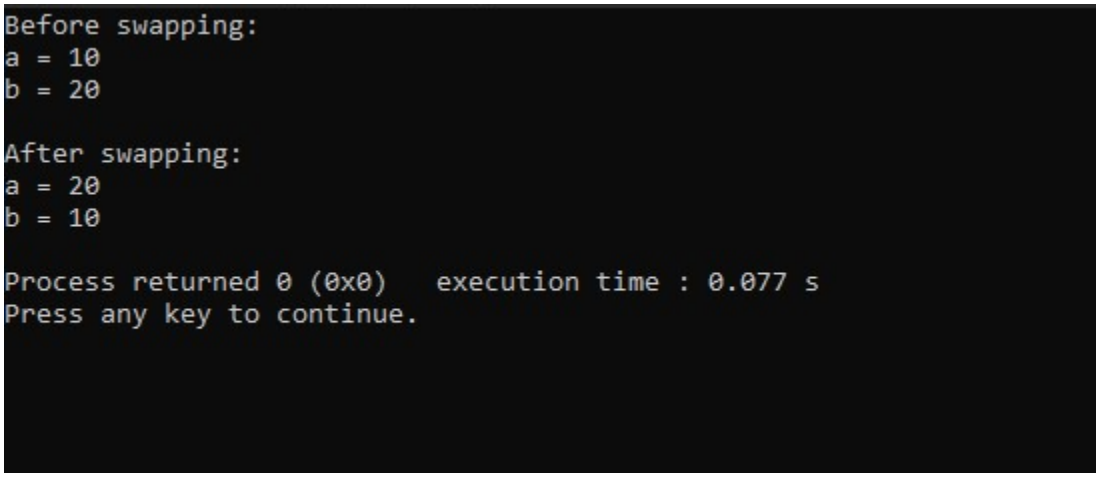
    printf("\nAfter swapping:\n");
    printf("a = %d\n", a);
    printf("b = %d\n", b);

    return 0;
}
```

Explanation

- Two integer variables, `a` and `b`, are initialized with values 10 and 20.
- The variable `temp` temporarily stores the value of `a`.
- The value of `b` is assigned to `a`.
- The original value of `a`, stored in `temp`, is assigned to `b`.
- The program displays the values before and after swapping.

Output



Result

The program successfully swaps the values of two integer variables using a temporary variable.

Task 3: Store and Display the Number and Price of Books

Objective

To write a C program that declares an integer variable for the number of books and a float variable for the price of a book, then displays both values.

Source Code

```c
#include <stdio.h>

int main() {
    int numberOfBooks = 5;
    float priceOfBook = 250.50;

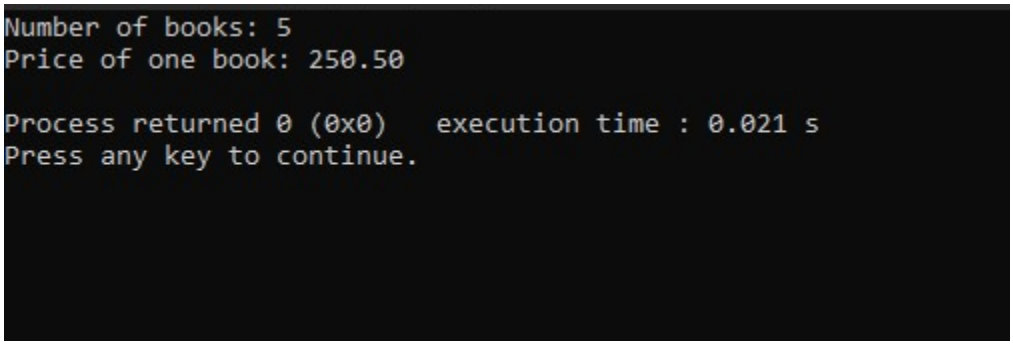
    printf("Number of books: %d\n", numberOfBooks);
    printf("Price of one book: %.2f\n", priceOfBook);

    return 0;
}
```

Explanation

- `int numberOfBooks` stores the number of books as a whole number.
- `float priceOfBook` stores the price of one book, including decimal values.
- The integer format specifier `%d` displays the number of books.
- The format specifier `%.2f` displays the price with two digits after the decimal point.
- The `printf()` function displays both values.

Output



Result

The program successfully declares an integer and a float variable and displays their values.